Lufan Yang

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PROFESSIONAL SUMMARY

I am a third-year Ph.D. student in Animal Sciences at the University of Illinois Urbana-Champaign, advised by Dr. Terence Crofts. My current research uses the Crofts Lab's in-house functional metagenomics platform to construct and screen environmental DNA libraries for novel soil-derived genes that confer resistance to streptothricin/nourseothricin (NTC). I focus on determining whether candidate resistance phenotypes are mediated by acetyltransferases, methyltransferases, or hydrolases through functional screening, sequence analysis, and experimental validation. My prior work integrated microbiome and transcriptomic analyses to investigate influenza A virus pathogenesis and maternal vaccination in neonatal piglets, providing interdisciplinary experience in antimicrobial resistance, microbial genomics, host-microbe interactions, and computational biology.

EDUCATION

Ph.D. in Animal Sciences, University of Illinois Urbana-Champaign, IL, USA — **Aug. 2024 - Present**
Advisor: Dr. Terence Crofts

M.S. in Pathobiology, University of Illinois Urbana-Champaign, IL, USA — **Aug. 2022 - May 2024**

Bachelor of Veterinary Medicine, Nanjing Agricultural University, Nanjing, China — **Sept. 2017 - July 2022**
GPA: 3.85 / 4.5

PROFESSIONAL EXPERIENCE

Researcher, Key Laboratory of Animal Virology, Ministry of Agriculture, China — **Oct. 2021 - Jan. 2022**

Participant, Online Summer School "From Farm to Fork", Wageningen University, Netherlands — **Aug. 2021**

PUBLICATIONS

1. **Yang, L.***, Lindahl, A.*, & Robben, M. (2026). Immunogenetics and TCR expression patterns together predict future T1D onset from multi-omic datasets. *Frontiers in Genetics*, 17, 1833788. doi:10.3389/fgene.2026.1833788
2. Gaulke, C. A., Yuan, F., **Yang, L.**, Duan, L., Connolly, M. G., Hsiao, S.-H., Antonson, A. M., & Fang, Y. (2025). Maternal vaccination partially protects piglets against influenza A virus associated alteration of the microbiome and hippocampal gene expression. *Veterinary Microbiology*, 306, 110544. doi:10.1016/j.vetmic.2025.110544
3. Yuan, F.*, **Yang, L.***, Hsiao, S.-H., Herndon, N. L., Gaulke, C. A., & Fang, Y. (2024). A neonatal piglet model reveals interactions between nasal microbiota and influenza A virus pathogenesis. *Virology*, 592, 109996. doi:10.1016/j.virol.2024.109996

* Equal contribution.

CONFERENCE ATTENDANCE

Lufan Yang et al., Effects of maternal vaccination on hippocampal transcriptome and nasal microbiome in neonatal piglets infected by Influenza A virus. 2023 NAPRRS/NC229 (Oral & Poster)

Lufan Yang et al., A neonatal piglet model reveals interactions between nasal microbiota and influenza A virus pathogenesis. 2024 Mining Microbial Genomes Meeting (Oral)

HONORS & AWARDS

NA PRRS International Conference Travel Award, Chicago, IL	**2023**
University Scholarship	**2023**
First-Class Scholarship for Three Good Students (top 3%)	**2021**
Second-Class Scholarship for Three Good Students (top 5%)	**2021, 2020, 2019, 2018**
Inspirational Scholarship	**2020**